

The River Court Filing Drift

How a court falls behind, how it catches up, and the eight equations that show it

$$CR_t = D_t \div F_t$$

cases finished ÷ cases filed

$$B_{t+1} = B_t + F_t - D_t$$

next pile = this pile + what came in – what went out

The question on the cover: did the court finish at least as many cases as came in?

Prepared for Key · Edition 639 · Legal information, not legal advice
Authors are EdereAirah staff of the Court Operations & Access Desk (simulated persons; no Earth credential claimed)

How to read every equation in this report

Each equation is shown whole first, so the eye learns its shape the way it knows 2×2 . Then each part is named, said in words, swapped for everyday things, worked with numbers, and closed with the one question it answers. Once you have seen it six ways, the symbols alone are enough: you look at $D \div F$ and you already know “finished over filed.”

Step	What you do	Example with $CR = D \div F$
1 · SEE	Look at the whole formula before reading anything	$CR = D \div F$
2 · NAME	Each letter gets one meaning, written right beside it	$D =$ cases finished, $F =$ cases filed
3 · SAY	Read it as a sentence	“The rate is what we finished divided by what came in.”
4 · SWAP	Put your own things in the letters; the shape still works	Dishes washed $\div$ dishes used
5 · WORK	Put real numbers in	$1,020 \div 1,200 = 0.85$
6 · ASK	Name the question the formula answers	Are we keeping up?

Truth labels used throughout

Label	Meaning
EARTH SOURCE	A published Earth figure, cited on page 14
WORLD SIMULATION	Numbers from the EdereAirah River Court story. Not Earth data, not measured anywhere
UNKNOWN	Not known yet. Left visible instead of guessed

1 • The problem: the River Court Filing Drift

Written by Tamsin Rourke, Court Systems Analyst (EdereAirah, simulated). Reviewed by Judge Mireille Oduya, Presiding Referee, Tenancy & Small-Claims Bench (EdereAirah, simulated; 11 years on the bench).

On the Tenancy & Small-Claims bench, the Filing Hall finished fewer cases than it received for four straight quarters (a quarter is 90 Earth days). About one hearing in four was postponed because the file was not ready. About one person in five did not appear. The pile grew every quarter.

No single person was failing. The steps did not check each other.

The seven steps a case walks through

#	Step	Who	Where it broke
1	Filing	The person filing, often without a lawyer	Missing names, addresses, attachments
2	Intake	Counter clerk	No completeness check: incomplete files went straight on
3	Scheduling	Docket supervisor	Hearing dates given to files that weren't ready
4	Hearing	Judge	Missed appearances; unready files postponed
5	Order	Judge	Orders people could not act on
6	Follow-through	Access desk + clerk	Nobody owned what happens after
7	Record	Archive custody	Files closed without a disposition code

2 • What we changed

Change	Owner (simulated)	What it does
Completeness check at intake	Rosalind Petrakis, Chief Deputy Clerk	One-page checklist; incomplete files go back the same day with a list
Ready-to-hear gate	Rosalind Petrakis	No hearing date until intake says COMPLETE
Plain-language notice	Naya Aven, Director of Access	Date, time, place and what happens if you miss it, printed largest
Reminder + “what you must do next”	Naya Aven	A reminder before the hearing; one next-step line on every order
Weekly clearance ledger	Hollis Varga, Court Systems Engineer	All eight measures on one sheet, every week

3 · Before and after

WORLD SIMULATION. Averages per 90-day quarter over four quarters. Filings held at 1,200 so the change can be seen on its own.

Measure	Before	After	Change
Filings F	1,200	1,200	held constant
Cases finished D	1,020	1,380	+360
Clearance rate $CR = D \div F$	0.85	1.15	+0.30
Pile change per quarter ($F - D$)	+180	-180	reversed
Backlog B over 4 quarters	2,400 → 3,120	3,120 → 2,400	back to start
Missed appearances (of 1,500)	330	264	-66
Failure-to-appear rate	22.0%	17.6%	-20% relative
Postponed, file not ready	405 (27%)	180 (12%)	-225 hearings
Median days to finish	140	120	-20 days
Slowest 10% (P90) days	410	340	-70 days
Clerk hours per finished case	2.1	1.55	-26%

How the numbers check each other

Clerk capacity $2,160 \text{ hours} \div 2.1 \text{ hours per case} = 1,029$, which matches 1,020 before. $2,160 \div 1.55 = 1,394$, enough for 1,380 after (M6). Freed hearing slots: $225 + 66 = 291$ per quarter (M3, M7). Mean wait at the end: $2,400 \div 13.33 \text{ per day} = 180 \text{ days}$ (M5).

4 · The eight equations

M1 · Clearance rate

DEFINITION · target ≥ 1.00

1 · SEE IT

$$CR_t = D_t \div F_t$$

say it aloud: “C-R sub t equals D sub t divided by F sub t”

2 · NAME EVERY PART

symbol	what it is	unit
CR_t	clearance rate in period t	ratio
D_t	dispositions: cases finished and leaving the court	cases
F_t	filings: new and reopened cases coming in	cases
t	the period: a month or a 90-day quarter	time

3 · SAY IT IN WORDS

Did the court finish at least as many cases as came in? 1.00 means keeping pace. Below 1.00, the pile grows.

5 · WORK IT

Earth (NCSC example): $913 \div 1,083 = 0.84$.

World before: $1,020 \div 1,200 = 0.85$.

World after: $1,380 \div 1,200 = 1.15$.

4 · SWAP IN YOUR OWN THINGS

Dishes washed $\div$ dishes used. Emails answered $\div$ emails received. The shape works the same with any “out $\div$ in.”

FOR A CHILD

If 10 new toys land in the box and you put away 8, the box gets fuller.

6 · THE QUESTION IT ASKS

Are we finishing as fast as cases arrive?

M2 · Backlog (the pile)**COUNTING IDENTITY****1 · SEE IT**

$$B_{t+1} = B_t + F_t - D_t$$

say it aloud: “B sub t-plus-one equals B sub t plus F sub t minus D sub t”

2 · NAME EVERY PART

symbol	what it is	unit
B_t	cases waiting at the start of period t	cases
B_{t+1}	cases waiting at the start of the next period	cases
F_t	cases that came in during t	cases
D_t	cases that left during t	cases

3 · SAY IT IN WORDS

The pile at the end is the pile at the start, plus what came in, minus what went out.

5 · WORK IT

Before: $2,400 + 1,200 - 1,020 = 2,580$; four quarters later 3,120.
 After: $3,120 + 1,200 - 1,380 = 2,940$... back to 2,400.

4 · SWAP IN YOUR OWN THINGS

Water in a bucket = old water + poured in – poured out. Money in the bank works the same way.

FOR A CHILD

A bucket fills when you pour in faster than it drains.

6 · THE QUESTION IT ASKS

Is the pile getting bigger or smaller, and by how much?

M3 · Failure to appear, and the reminder effect**DEFINITION + EARTH-TESTED
MODEL****1 · SEE IT**

$$\text{FTA} = M \div S ; \text{FTA}_{\text{after}} = \text{FTA}_{\text{before}} \times (1 - r)$$

say it aloud: “F-T-A equals M over S; F-T-A after equals F-T-A before times one minus r”

2 · NAME EVERY PART

symbol	what it is	unit
M	hearings where a required person did not come	hearings
S	hearings scheduled that needed someone to appear	hearings
r	how much a reminder cuts misses, as a fraction (0.20 = 20%)	ratio

3 · SAY IT IN WORDS

The share of people who didn't come, and how much a reminder shrinks that share.

5 · WORK IT

Before: $330 \div 1,500 = 0.22$.

After: $0.22 \times (1 - 0.20) = 0.176 \rightarrow 264$ of 1,500.

Earth: reminders cut misses 13-21% in NYC (Science, 2020).

4 · SWAP IN YOUR OWN THINGS

Missed doctor visits $\div$ booked visits. Practice no-shows $\div$ practices called.

FOR A CHILD

If 5 friends are invited and 1 forgets, a note might mean nobody forgets.

6 · THE QUESTION IT ASKS

How many people miss court, and what brings them back?

M4 · Time to finish (median and slow tail)**DEFINITION · standards are benchmarks****1 · SEE IT**

$$T_i = \text{date}_{\text{finished}} - \text{date}_{\text{filed}} ; S(\tau) = \text{count}(T_i \leq \tau) \div N$$

say it aloud: “T sub i equals date finished minus date filed; S of tau equals how many are at most tau, over N”

2 · NAME EVERY PART

symbol	what it is	unit
T_i	days case i was open	days
τ (tau)	the time standard being checked, e.g. 90 days	days
N	cases finished in the period	cases
$S(\tau)$	share of cases finished within τ days	ratio
P_{50}, P_{90}	median case, and the slowest 10% line	days

3 · SAY IT IN WORDS

How long cases take, for the typical case and for the slow ones that get forgotten.

5 · WORK IT

World: median 140 → 120 days; P_{90} 410 → 340 days.

Earth misdemeanor standard: $S(60) \geq 0.75$, $S(90) \geq 0.90$, $S(180) \geq 0.98$.

4 · SWAP IN YOUR OWN THINGS

How long orders take to ship: the typical order, and the slowest one in ten.

FOR A CHILD

Line everyone up by how long they waited. The middle person is the median.

6 · THE QUESTION IT ASKS

How long does a person wait for an answer?

M5 · Little's law (average wait)**QUEUEING LAW (steady state)****1 · SEE IT**

$$W = L \div \lambda$$

say it aloud: "W equals L divided by lambda"

2 · NAME EVERY PART

symbol	what it is	unit
W	average time a case spends waiting	days
L	average number of cases waiting	cases
λ (lambda)	how fast new cases arrive	cases/day

3 · SAY IT IN WORDS

A big pile with a slow arrival rate means each case waits a long time.

5 · WORK IT

$\lambda = 1,200 \div 90 = 13.33$ per day.
 $W = 2,400 \div 13.33 = 180$ days.

4 · SWAP IN YOUR OWN THINGS

Cars in a car-wash line $\div$ cars arriving per minute = minutes you'll wait.

FOR A CHILD

20 kids in line, 2 new ones each minute: each waits about 10 minutes.

6 · THE QUESTION IT ASKS

If you join the line today, how long until your turn?

M6 · Clerk capacity**WORKLOAD MODEL****1 · SEE IT**

$$D_{\max} = (H \times u) \div \bar{h}$$

say it aloud: "D max equals H times u, divided by h-bar"

2 · NAME EVERY PART

symbol	what it is	unit
H	clerk hours available in the period	hours
u	share of those hours spent on case work	ratio
$\bar{h}$ (h-bar)	average clerk hours per finished case	hours
$D_{\max}$	most cases the clerks can finish	cases

3 · SAY IT IN WORDS

Hours you have, times the share you can use, divided by hours per case, is the most you can finish.

5 · WORK IT

6 clerks $\times$ 450 h = 2,700 h; $\times$ 0.8 = 2,160 h.
Before: $2,160 \div 2.1 = 1,029$. After: $2,160 \div 1.55 = 1,394$.

4 · SWAP IN YOUR OWN THINGS

Baking: oven hours $\times$ share used $\div$ hours per batch
= batches you can make.

FOR A CHILD

Each puzzle takes 2 hours and you have 10 hours:
you finish 5 puzzles.

6 · THE QUESTION IT ASKS

Can the people we have keep up with what comes in?

M7 · Wasted hearings**MODEL · target ready share ≥ 0.85** **1 · SEE IT**

$$X = S \times p_c$$

say it aloud: "X equals S times p sub c"

2 · NAME EVERY PART

symbol	what it is	unit
X	hearing slots used up with no progress	hearings
S	hearings scheduled	hearings
p_c	share postponed because the file wasn't ready	ratio

3 · SAY IT IN WORDS

How many hearing slots were spent going nowhere.

5 · WORK IT

Before: $1,500 \times 0.27 = 405$. After: $1,500 \times 0.12 = 180$.
225 slots recovered every quarter.

4 · SWAP IN YOUR OWN THINGS

Delivery trucks sent out $\times$ share that arrived with the wrong box.

FOR A CHILD

Showing up to a game with no ball wastes the field.

6 · THE QUESTION IT ASKS

How much court time is lost to files that weren't ready?

M8 · Pilot decision rule**DECISION RULE****1 · SEE IT**

$$\Delta = (x_{\text{after}} - x_{\text{base}}) \div x_{\text{base}}$$

say it aloud: "Delta equals x after minus x base, over x base"

2 · NAME EVERY PART

symbol	what it is	unit
Δ (delta)	relative change	ratio
x_{base}	the measure before the change	any
x_{after}	the same measure after the change	same

3 · SAY IT IN WORDS

Did it move enough? PASS only if it meets the target in at least 2 of 3 pilot months.

4 · SWAP IN YOUR OWN THINGS

Grocery bill this month vs last month, as a percent.

5 · WORK IT

FTA 0.22 → 0.19: $\Delta = (0.19 - 0.22) \div 0.22 = -0.136$. Meets the 10% target.

FOR A CHILD

Did your score go up in at least two of three games?

6 · THE QUESTION IT ASKS

Did our change really work, or was it luck?

5 · A 90-day pilot any court can run

Field	Plan
Partner	UNKNOWN: no court contacted yet. Target: one docket (small claims or landlord-tenant)
Data	Aggregate monthly counts only. No names, case numbers or case facts, ever
What changes	Intake checklist · ready-to-hear gate · plain notice · reminder through the court's own channel
Pass if	CR $\geq$ 1.00 in 2 of 3 months · FTA drop $\geq$ 10% · ready share $\geq$ 0.85 · slowest-10% time not worse
Stop if	Any privacy issue · anyone says the checklist blocked a filing (filings are flagged, never refused)
Undo	Paper checklist and template. Stop using the sheet and it's gone

6 · Privacy line we never cross

No personal details of anyone in a case. Sealed, juvenile, protective-order, mental-health, adoption and expunged matters are excluded completely. Any count below 11 is hidden. Reminders are sent only by the court, under its own rules. We never hold a contact list.

7 · Who wrote this (simulated EdereAirah staff)

Person	Title	Section
Nadia Baptiste	Director, Global Legal Operations	Department lead
Tamsin Rourke	Court Systems Analyst (proposed)	Story, results, math
Judge Mireille Oduya	Presiding Referee, Tenancy & Small-Claims Bench (proposed)	Bench review
Rosalind Petrakis	Chief Deputy Clerk, The Filing Hall (proposed)	Workflow
Naya Aven	Director of Access and Follow-Through	Notices, reminders
Hollis Varga	Court Systems & Software Engineer (proposed)	Ledger, dashboard
Elena Marrow	Privacy, Data & Consumer Protection	Privacy

All are EdereAirah persons. No Earth credential is claimed for any of them.

8 · Earth sources

OPENED = page read and quoted. SNIPPET = seen in search results only; recheck before publishing.

#	Source	Figure used	Level
S1	NCSC CourTools Measure 2: Clearance Rates	Definition; $\geq 100\%$ target; $913 \div 1,083 = 84\%$	OPENED
S2	NCSC CourTools Measure 3: Time to Disposition	Definition	SNIPPET
S3	Model Time Standards for State Trial Courts (2011)	Misdemeanor 75%/60d, 90%/90d, 98%/180d	OPENED
S4	Fishbane, Ouss & Shah, Science 370 (2020)	FTA -13% to -21% ; $\sim 30,000$ fewer warrants in 3 yrs	OPENED
S5	NCSC Landscape of Civil Litigation (2015)	76% of civil cases had a self-represented party	SNIPPET
S6	Court Statistics Project (2022 data)	64.6 million incoming cases	SNIPPET
S7	LSC, The Justice Gap (2022)	92% of civil legal problems got little or no help	SNIPPET
S9	Thomson Reuters Institute 2025 State Courts Survey	Clerk shortages expected to continue	OPENED

9 · Print check (run this page first)

If every symbol below prints as itself, the printer and fonts are good. A box □ or blank means stop and tell us.

Symbol	Name	Symbol	Name
×	times	÷	divided by
–	minus	$\leq \geq$	at most / at least
Σ	sum of	Δ	change
λ	lambda	τ	tau
$\bar{h}$	h-bar (average)	$\rightleftharpoons$	goes both ways
B_{t+1}	subscript	x^2	superscript

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